Subscriber Churn Prediction & Next-Best-Action Analysis

A predictive model flags which of 5,000 subscribers are likely to cancel, a segmentation layer groups the at-risk ones by the behavior driving that risk, and a decision framework prices out a specific save for each group — in place of one blanket discount for everyone.

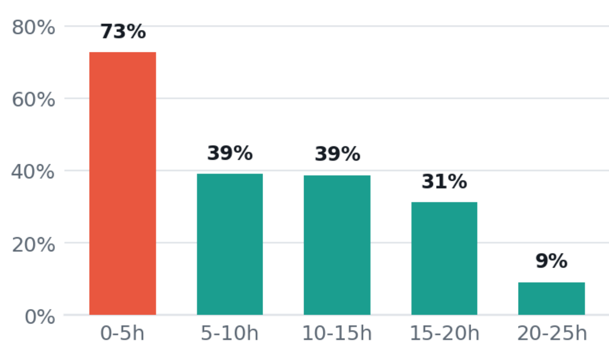
5,000	0.88	2,874	$85,326
Subscribers analyzed	Model accuracy · ROC-AUC	At-risk subscribers	Est. annual savings

01 · ROOT CAUSES

Why subscribers leave

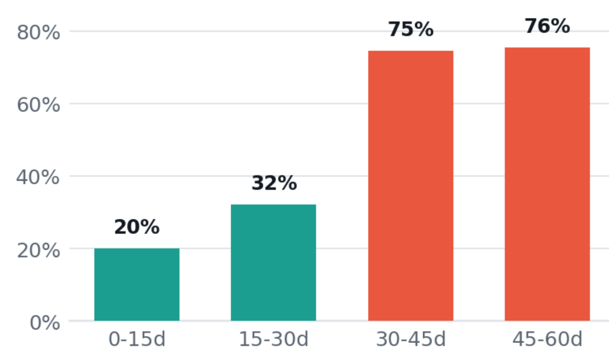
Two behaviors predict churn far more than anything else. Demographics — age, gender, region, device — are noise.

Churn by weekly watch hours — r = –0.54



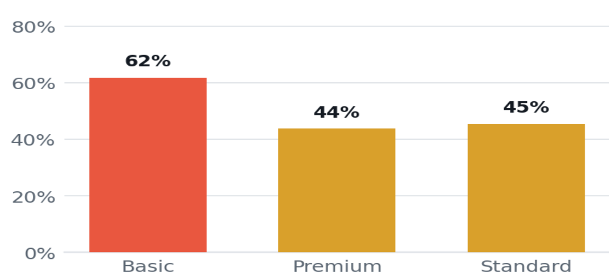
Under 5 hrs/week churns at **73%**; past 20 hours, that falls to **9%**.

Churn by days since last login — r = +0.72



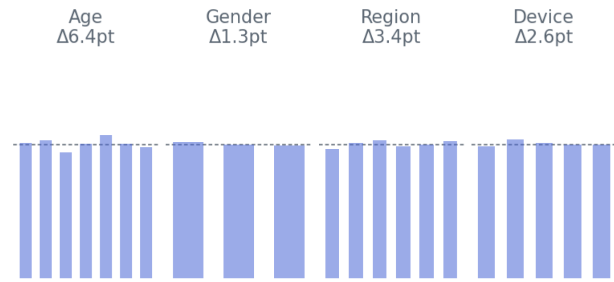
Churn nearly quadruples once a subscriber's been quiet 30+ days — the strongest predictor in the model.

Churn rate by plan



Basic churns at roughly 1.4× the rate of Standard or Premium — price alone doesn't explain it.

What doesn't move the needle — all within ~7pts of a coin flip



Age, gender, region, device: none meaningfully separate stayers from leavers.

02 · SEGMENTATION

Three ways to lose a subscriber

K-Means grouped the 2,874 at-risk subscribers into three behaviorally distinct segments. Each gets a different save, sized to what they're worth and why they're drifting.

Higher-value, going quiet

907 subscribers · 7.45 avg hrs/wk · 50.6 avg days quiet · $15.99/mo avg

Root cause — Pays above-average fees, but has gone a median of 50 days without logging in — the longest quiet streak of any segment.
Action — Proactive win-back outreach: one month free, or a complimentary plan upgrade.

$34,366
value at risk recovered

Basic plan, going quiet

1,185 subscribers · 5.99 avg hrs/wk · 36.6 avg days quiet · $8.99/mo avg

Root cause — On the lowest-cost plan already, with a median of 37 days since last login. Price isn't the lever here.
Action — Low-cost reactivation: automated reminder email or push, with a free-week offer.

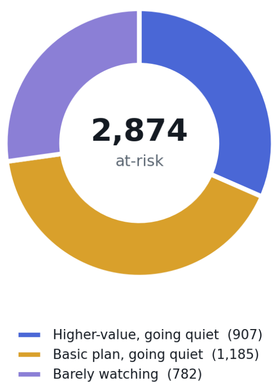
$25,568
value at risk recovered

Barely watching

782 subscribers · 2.78 avg hrs/wk · 30.0 avg days quiet · $15.91/mo avg

Root cause — Engagement has collapsed to under 3 hours watched — they haven't cancelled yet, but they've effectively stopped using the product.
Action — Content-led re-engagement: a personalized, genre-based recommendation email.

$25,392
value at risk recovered



Assumption set. A targeted intervention recovers **25%** of subscribers otherwise expected to churn within a segment. Retained-subscriber value is calculated as **12 months** of that subscriber's monthly fee. This assumption is stated explicitly so it can be adjusted against real historical win-back rates — the $85,326 total scales directly with it.

Total estimated annual savings: $85,326, summed across all three segments.

03 · HOW THIS WAS BUILT

Methodology

DATA	MODEL	SEGMENTS	DECISION LAYER
5,000 subscribers cleaned and structured across watch time, login recency, plan tier, monthly fee, and account tenure indicators.	Logistic Regression estimates churn probability per subscriber — chosen for coefficient interpretability alongside strong performance (ROC-AUC 0.88, recall 0.84).	K-Means clusters the 2,874 medium- and high-risk subscribers into three behavioral groups by engagement, plan tier, and login recency.	Each segment maps to one retention action, priced with a stated, adjustable assumption set rather than a single blanket offer.

04 · SUMMARY

Key findings

Engagement is the dominant driver. Subscribers watching under 5 hours a week churn at 73%, versus 9% for those watching 20+ hours — by far the widest swing in the data.	**Plan tier compounds risk independently.** Basic-plan subscribers churn at roughly 1.4× the rate of Standard or Premium, even after accounting for engagement.
Recency and engagement outrank demographics. Login recency and watch time are the model's two strongest predictors; age, gender, and region show no meaningful relationship to churn.	**At-risk subscribers aren't one population.** Three distinct segments emerge, each needing a different intervention rather than a single uniform offer.

TOOLS Python (pandas for data prep, scikit-learn for classification and clustering) for the analytical pipeline · Tableau for the stakeholder-facing dashboard · companion interactive HTML console for live filtering.

DATASET 5,000-subscriber streaming platform sample: subscription tier, watch hours, login recency, monthly fee, and churn outcome.